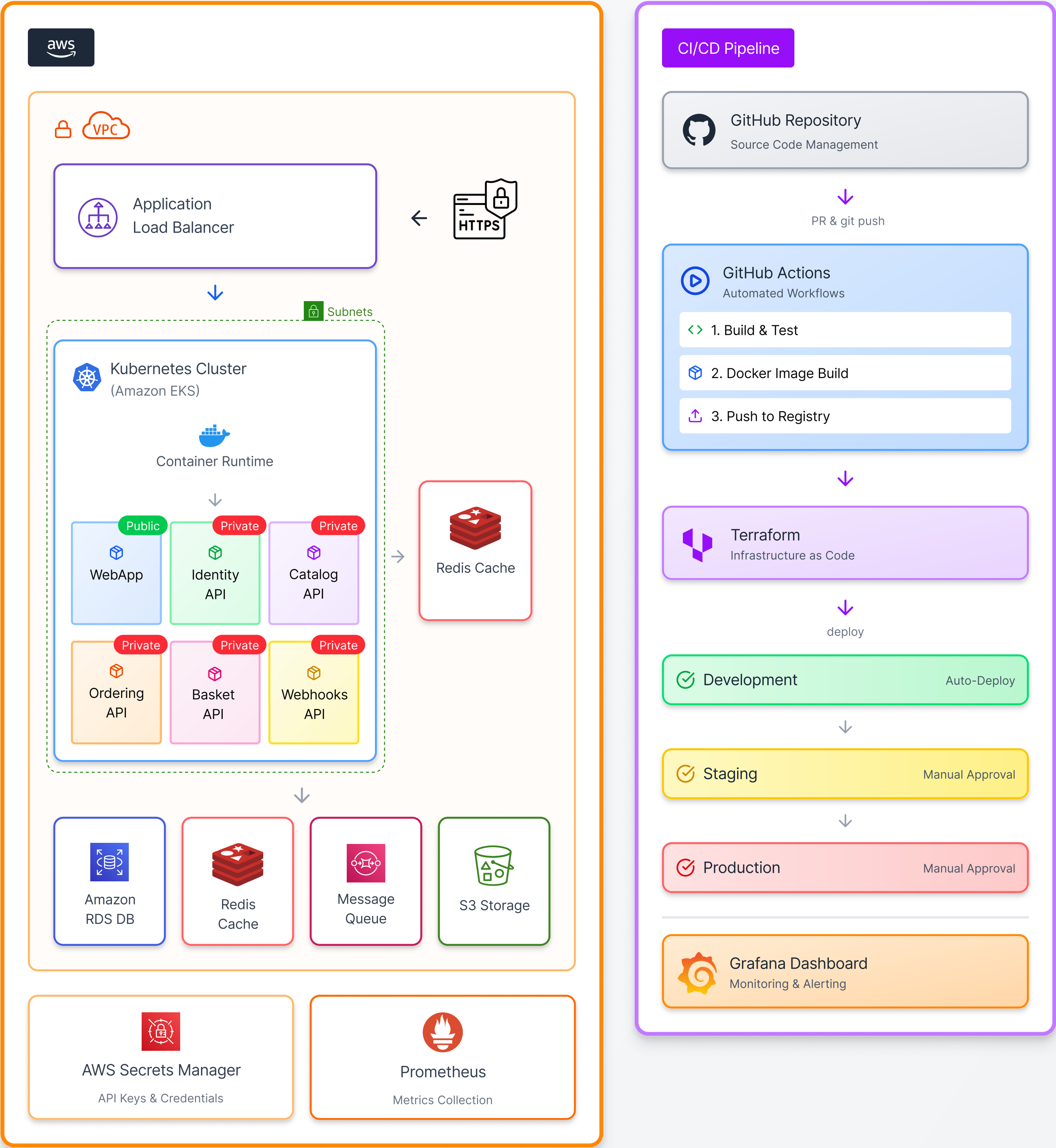


System Architecture: AWS Cloud Infrastructure & CI/CD Pipeline

Kubernetes-based microservices architecture with automated deployment pipeline

Amazon EKS • Docker Container • PostgreSQL • Redis • RabbitMQ • S3 • Application Load Balancer • Prometheus Monitoring • Grafana Dashboards



AWS Infrastructure

The microservices run in a Kubernetes cluster (EKS) within a VPC. The Application Load Balancer routes traffic to the various services. Databases and message queues support persistence and communication.

CI/CD Pipeline

GitHub Actions automates build, test and deployment. Terraform manages the infrastructure. Code is deployed through Development → Staging → Production. Grafana monitors Prometheus metrics.